

Database Concepts

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We store data in Files, A file can be understood as a container to store data in a computer. Files can be stored on the storage device of a computer system. Content of a file can be texts, computer program code, comma separated values(CSV), etc. Likewise pictures, audios/videos, web pages are also files.

Files stored on a computer can be accessed directly and search for desired data. But to access data of a file through software, for example to display monthly attendance report on a school website. One has to write computer program to access data from files.

Student File which has 6 columns:

RollNumber - Roll number of a student.

Sname - Name of the student.

SDateOfBirth - Date of birth of the student.

Gname - Name of the Guardian.

Gmobile - Mobile number of the students guardian.

Gaddress- Address of the guardian of the student.

Another file which is updated Regularly called Attendance which has 4 columns:

AttendanceDate - Date for which attendance was Marked.

RollNumber - The roll number of the student.

Sname - Name of the Student

AttendanceStatus - Marked as P (present) or A(absent)

Limitations of a file system

File systems become difficult to handle when number of files increases and volume of data also grows.

Difficulty in Access: Files themselves do not provide any mechanism to retrieve data. Data maintained in a file system are accessed through application programs. While writing such programs, the developer may not anticipate all the possible ways in which data may be accessed. So sometimes it is difficult to access data in the required format and one has to write application program to access data.

Data Redundancy: Redundancy means same data are duplicated in different places (files) In our example, student names are maintained in both the files. Students may have. Same guardian name and therefore same guardian name is maintained twice. This is the example of redundancy which is difficult to avoid in a file system. Redundancy leads to excess storage use and may cause **data inconsistency** also.

Data inconsistency: Data inconsistency occurs when same data maintained in different places do not match. If a student wants to get changed the spelling of her name. It needs to be changed in Sname column in both the files. Likewise, if a student leaves school, the details needs to be deleted from both the files. As the files are being maintained by different people, the changes may not happen in one of the files. In that case the student name will be different (inconsistent) in both the files.

Data Isolation: Both the files student and attendance are related to

students. But there is no link or mapping between them. The school will have to write separate programs to access these two files. This is because data mapping is not supported in file system. In a more complex system. Data files are generated by different persons at different times. Files being created in isolation may be of different formats. In such case it is difficult to write new application program to retrieve data from different files maintained at multiple places as one has to understand the underlying structure of each file as well.

Data Dependence: Data are stored in a Format or structure in a file. If the structure or format itself is changed, all the existing application programs accessing the file also needs to be changed. Otherwise the program may not work correctly. This is data dependency. Hence, updating the structure of a data file requires modification in all the application program accessing that file.

Controlled Data Sharing: There can be different categories. Categories of users like teacher, office, staff and parents. Ideally, not every user should be able to access all the data. As an example, guardian and office staff can only see the student attendance data, but should not be able to modify or delete it. It means these users should be given limited access (read only) to the attendance file. Only the teacher should be able to update the attendance data. It is very difficult to enforce this kind of access control in a file system where accessing file through application programs.

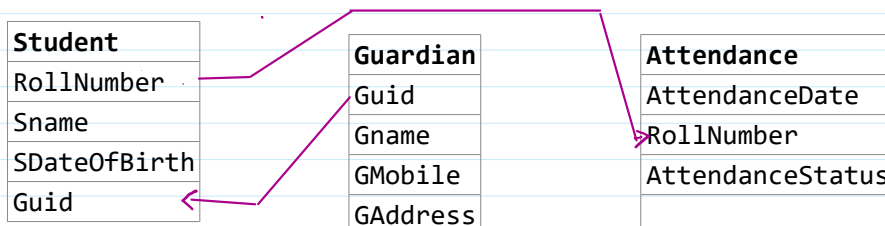
DATABASE MANAGEMENT SYSTEM

Limitation faced in the file system can be overcome by storing the data in a database where data are logically related. We can organize related data in a database so that it can be managed in an efficient and easy way.

A database management system (DBMS) or database system in short is a software that can be used to create and manage databases. DBMS let's user to create a database store, manage, update, or modify and retrieve data from the database by users or application programs. Some examples of open source and commercial DBMS include MySQL, Oracle PostgreSQL, , Microsoft SQL server, MongoDB, DB2 by IBM etc.

A database system hides certain details about how data are actually stored and maintained. Thus it provides users with an abstract view of the data. A database system has a set of programs through which users or other programs can access, modify, and retrieve the stored data.

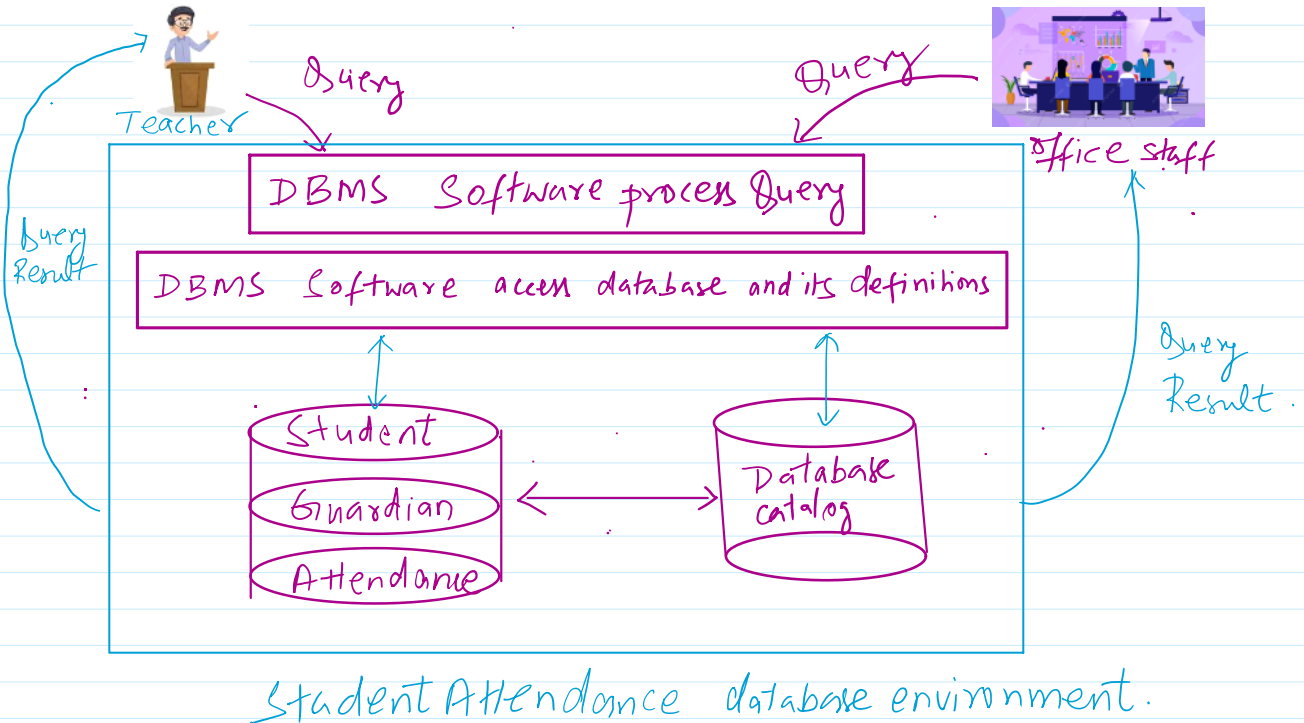
The DBMS serves as an interface between the database and end users or application programs. Retrieving data from a database through special type of command is called querying the database. In addition, user can modify the structure of the database itself through a DBMS.



Key Concepts in DBMS

In order to efficiently manage data using a DBMS:

Database Schema: Database schema is the Design of a database. It is the skeleton of a database that represents the structure (table names and their fields or columns) The type of data each column can hold. Constraints on the data to be stored (if any) And the relationship among the tables. Database schema is also called the visual or logical architecture as it tells how the data are organized in a database.



What is DBMS?

A DBMS refers to a software that is responsible for storing, maintaining and utilizing databases. A database along with a DBMS is referred to as a database system.

What is data redundancy?

Duplication of data is known as Data Redundancy.

What is data inconsistency?

Mismatched multiple copies of same data, is known as Data Inconsistency.

Hostel(reg_num, name, room_no, floor_no, mobile_no, permanent_add)
(This is not updated)

Student(reg_num, name, sem, course, mobile_no, email_id, address)
This is updated

Relational Database Management System

In relational data model, the data is organized into tables (rows and columns). These tables are called relations. A row in a table represents a relationship among a set of values. Since a table is a collection of such relationships, it is generally referred to using the mathematical term relation, from which the relational data model derives its name.

Let us see how a sample database can be represented in relational form. The sample database has three tables(relation): Suppliers, Items, Shipments

Suppliers(**SuppNo**, SuppName, Status, City)

Items(**ItemNo**, ItemName, Price)

Shipments(**SuppNo**, **ItemNo**, Qty_supplied)[Bold fields represents primary keys]

Primary key is the key field that can uniquely identify a row in a relation.

Each of these tables closely resembles a conventional sequential file, with rows of a table corresponding to records of a file and columns corresponding to the fields of the record. Each of these table is actually a special case of the construct known in mathematics as a relation.

Rows of relation are generally referred to as **tuples**, and columns are usually referred to as **attributes**.

Suppliers

suppNo	Supp_Name	Status	City
S1	Britannia	10	Delhi
S2	New Bakers	30	Mumbai
S3	Mother Dairy	10	Delhi
S4	Cookz	50	Bangaluru
S5	Haldiram	40	Jaipur

Items

ItemNo	ItemName	Price
I1	Milk	35.00
I2	Cake	5.00
I3	Bread	40.00
I4	Milk bread	50.00
I5	Plain Biscuit	10.00
I6	Cream Biscuit	20.00
I7	Ice Cream	60.00
I8	Cold Drink	20.00
I9	Namkeen	15.00
I10	Cookies	100.00

Primary key

← records

← data item or elements

Shipments

Suppno	ItemNo	Qty_supplied
S1	I2	10
S1	I3	20
S1	I6	20
S2	I4	20
S2	I5	10
S3	I1	10
S3	I7	10
S4	I8	30
S4	I9	30

Primary key

Foreign Key

Notice that here each supplier has a unique supplier number, exactly one name, status value and location. Likewise, we assume that each item has a unique item number, name, price, and also that at any given time, no more than one shipment exist for a given supplier or item combination.

Components of a table

These are:

Byte: A group of eight bits and is used to store a character.

Data Items(Field/attribute): A data item is the smallest unit of named data. It may consist of any number of bits or bytes. A data item represents one type of information and is often referred to as field or data element or attribute.

Record: A record is a named collection of data items which represents a complete unit of information.

Table or relation: A table is a named collection of all occurrences of a given type of a logical record.

Fields: Are the columns containing one type of information.

Examples of common relational database management system include Oracle Database, MYSQL, IMB DB2, IBM Informix, IBM Vector, SAP Sybase Adaptive Server Enterprise, SAP Sybase IQ Teradata, PostgreSQL, SQLite etc.

The Relational model terminology

The relational model was propounded by **Dr. E.F Codd** of the IBM and has since been acknowledged as a very important concept in DBMS technologies.

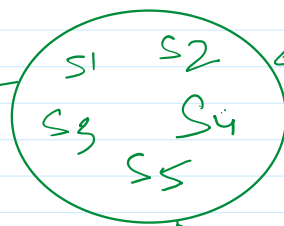
Relation: In general, a relation is a table that is data arranged in rows and columns. A relation has the following properties:

- i. In any given column of a table, all items are of the same kind, whereas item in different columns may not be of the same kind.
- ii. For a row, each column must have an atomic (indivisible) value. And also for a row, a column cannot have more than one value.
- iii. All rules of a relation are distinct. That is, a relation does not contain two rows which are identical in every column. That is, each row of a relation can be uniquely identified by its content.
- iv. There is no order maintained for rows inside a relation. That is, we cannot retrieve anything by saying that from row #5 column name to be accessed.
- v. The columns of a relation are assigned a distinct names and the ordering of these columns is immaterial.

Domain: A domain is a pool of values from which the actual values appearing in a given column are drawn. For example, the values appearing in the Supply number column of both the supplier table and the shipment tables are drawn from the underlying domain of all valid supplier numbers.

Shipments

Suppno	ItemNo	Qty_supplied
S1	I2	10
S1	I3	20



Supplier Number Domain.

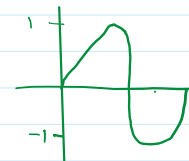
Suppliers

Suppno	ItemNo	Qty_supplied
S1	I2	10
S1	I3	20
S1	I6	20
S2	I4	20
S2	I5	10
S3	I1	10
S3	I7	10
S4	I8	30
S4	I9	30



Suppliers

suppNo	Supp_Name	Status	City
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S4	Cookz	50	Bangaluru
S5	Haldiram	40	Jaipur



$$\text{Domain of } \sin \theta = [-1, 1]$$

Terminology used in database software

- **Attribute:** Characteristics or parameters for which data are to be stored in the relation. Simply, Stated the columns of the relations are the attributes which are referred as fields. For example Suppno, ItemNo, Qty_Supplied are the attributes of the relation Shipments.
- **Tuple:** Each row of data in a relation (table) is called a tuple. In a table with N columns. A tuple is a relationship between N related values.
- **Domain:** It is a set of values from which an attribute can take a value in each row. Usually a data type is used to specify domain for an attribute. For example, Qty_supplied takes integer values and hence its domain is a set of integer values. Similarly, a set of character strings constitute the domain of the attribute Supp_name.
- **Degree:** The number of attributes in a relation is called a degree of the relation. For example, relation Suppliers have with four attributes is a relation of degree 4.
- **Cardinality:** The number of tuples in a relation is called the cardinality of the relation. For example. The cardinality of relation supplier is 5. As there are five tuples in the table.

Keys in a Relational Database

The tuples within a relation must be distinct. It means no tuples in a table should have same value for all attributes. That is, there should be at least one attribute in which data are distinct (unique) and not null. That way we can uniquely distinguish each tuple of a relation. So relational data model imposes some restriction or constraints on the values of the attributes and how the content of one relation be referred through another relation. These restrictions are specified at the time of defining the database through different types of keys as given below:

Candidate Key: A relation can have one or more attributes that takes a distinct values. Any of these attributes can be used to uniquely identify the tuples in the relation. Such attributes are called candidate keys, as each of them are candidates for the primary key.

Gname
GMobile
GAddress

The relation guardian has four attributes out of which Guid and Gmobile always takes unique values. No two guardians will have same phone number or mobile number or same Guid. Hence these two attributes are the candidate keys as they both are candidates for primary key.

Primary key: Out of one or more candidate keys, the attribute chosen by the database designer to uniquely identify the tuples in a relation is called primary key of that relation. The remaining attributes in the list of candidate keys are called the alternate keys.

In the relation guardian suppose GUID is chosen as primary key, then GMobile will be called the alternate key.

Composite Primary Key

If no single attribute in a relation is able to uniquely distinguish the tuples, then more than one attribute are taken together as primary key. Such primary key consist of more than one attribute is called composite primary key.

Suppno	ItemNo	Qty_supplied
S1	I2	10
S1	I3	20
S1	I6	20
S2	I4	20
S2	I5	10
S3	I1	10
S3	I7	10
S4	I8	30
S4	I9	30
S2	I2	20
S3	I6	40

Shipment

In relation Shipment, Suppno or ItemNo cannot be used as a primary key, as Suppno of same ItemNo will appear in another row, even Itemno cannot be used as primary key, as ItemNo of same Suppno will appear in another row. However combination of these two attributes Suppno and Itemno together would always have unique value in Shipment table or relation. Hence(Suppno, Itemno) will make the Shipment relation a composite primary key.

Foreign Key: A foreign key is used to represent the relationship between two relations. A foreign key is an attribute whose value is derived from the primary key of another relation. This means that any attribute of relation (referencing) which is used to refer contents from another (referenced) Relation becomes foreign key if it refers to the primary key of referenced relation. The referencing relation is called foreign relation. In some cases foreign key can take null value if it is not in the part of primary key of the foreign table. The relation in which the referenced primary key is defined is called primary relation or master relation.

